

I. PART - A

1. Answer **any ten** questions.

a) What do you mean by generations of computers?

Ans: Generations of computers refer to the classification of computers based on the technology used in their development.

b) Differentiate RAM and ROM.

Ans: RAM is volatile and temporary memory, whereas ROM is non-volatile and permanent memory.

c) What is Cache memory?

Ans: Cache memory is a high-speed memory that stores frequently used data for quick CPU access.

d) Define System Software. Give an example.

Ans: System software is software that manages and controls computer hardware, e.g., Operating System.

e) What are Language Translators?

Ans: Language translators are programs that convert high-level or assembly language into machine language.

f) What is Machine Language?

Ans: Machine language is the lowest-level language consisting of binary digits (0s and 1s).

g) Obtain the 1's and 2's Complements of the following binary numbers.

Ans: i) 1010101 $\rightarrow$ 1's complement: 0101010, 2's complement: 0101011

ii) 000001 $\rightarrow$ 1's complement: 111110, 2's complement: 111111

h) What is meant by RADIX of the number system?

Ans: Radix is the base or total number of digits used in a number system.

i) What is Excess-3 code? Give an example.

Ans: Excess-3 code is a binary code obtained by adding 3 to the decimal digit, e.g., $5 \rightarrow 1000$.

j) Write truth table and logic expression of NOR gate.

Ans:

k) What are Minterm and Maxterm?

Ans: Minterm is a product term that gives output 1, while maxterm is a sum term that gives output 0.

I) Write the general structure of 2 variable K-map.

Ans:

II. PART - B

2.

a) List and explain characteristics of Computers.

Ans: Characteristics of Computers

Computers have several important characteristics that make them powerful and useful:

- Speed: A computer can perform millions or billions of calculations per second, making it much faster than humans.
- Accuracy: Computers give highly accurate results if the input and program are correct.
- Diligence: A computer does not get tired or bored and can work continuously for long hours without errors.
- Storage Capacity: Computers can store a large amount of data and retrieve it whenever needed.
- Versatility: A computer can perform different types of tasks such as calculations, communication, gaming, designing, etc.
- Automation: Once a program is given, a computer works automatically without human intervention.

b) Draw the block diagram of a Computer System and explain the components of a Computer System.

Ans: Block Diagram of a Computer System and Its Components:

c) Briefly explain any two pointing device.

Ans:

- Mouse: A mouse is a pointing device used to move the cursor, select items, and perform actions by clicking.
- Touchpad: A touchpad is a flat pointing device found in laptops, used to control the cursor using finger movements.

3.

a) List and explain in brief the characteristics of Third-Generation Computers.

Ans:

- Integrated Circuits (ICs): Third-generation computers used ICs instead of transistors, which reduced size and cost.
- Higher Speed: They were much faster and could execute millions of instructions per second.
- More Reliable: ICs improved reliability and reduced hardware failures.
- Smaller Size: Computers became smaller and more compact compared to earlier generations.

- Lower Power Consumption: They consumed less power and produced less heat.
- Improved Operating Systems: Supported multiprogramming and time-sharing operating systems.

b) Explain the applications of Computers in different fields.

Ans:

- Education: Used for online learning, examinations, research, and digital libraries.
- Business: Used for accounting, payroll, inventory control, and data analysis.
- Healthcare: Used for patient records, medical imaging, diagnosis, and hospital management.
- Banking: Used for online banking, ATM services, and transaction processing.
- Science and Engineering: Used for research, simulations, designing, and complex calculations.
- Entertainment: Used for gaming, movies, music, animation, and content creation.

c) List out any four registers available in CPU along with its function.

Ans:

- Accumulator (ACC): Stores intermediate results of arithmetic and logical operations.
- Program Counter (PC): Holds the address of the next instruction to be executed.
- Instruction Register (IR): Stores the currently executing instruction.
- Memory Address Register (MAR): Holds the address of the memory location to be accessed..

4.

a) Define Operating System and explain basic functions of an Operating System.

Ans:

Definition: An Operating System (OS) is system software that acts as an interface between the user and computer hardware and manages all system resources.

Basic Functions of an Operating System:

- Process Management: Controls execution of programs and manages CPU time for different processes.
- Memory Management: Allocates and deallocates memory to programs and manages RAM efficiently.
- File Management: Organizes, stores, retrieves, and protects files on storage devices.
- Device Management: Controls and coordinates input and output devices like keyboard, mouse, printer, etc.
- Security and Protection: Prevents unauthorized access and protects data and system resources.

b) Write a note on:

Ans:

i) Word Processors: Word processors are application software used to create, edit, format, and print text documents. They provide features like spell check, text formatting, tables, images, and page layout.

Examples: Microsoft Word, Google Docs, LibreOffice Writer.

ii) Image Editors: Image editors are software used to create, edit, and modify digital images. They allow operations like cropping, resizing, color correction, and adding effects.

Examples: Adobe Photoshop, GIMP, Canva.

c) Write an algorithm to display maximum of three numbers.

Ans:

Step 1: Start

Step 2: Read three numbers A, B, and C

Step 3: If $A \geq B$ and $A \geq C$, then Max = A

Else if $B \geq A$ and $B \geq C$, then Max = B

Else Max = C

Step 4: Display Max

Step 5: Stop

5.

a) Briefly explain different phases of the Program Development Cycle.

Ans:

- Problem Definition: Understanding and clearly defining what the program is expected to do.
- Program Design: Planning the solution using algorithms and flowcharts.
- Coding: Writing the program using a suitable programming language.
- Testing and Debugging: Finding and correcting errors in the program.
- Documentation: Preparing manuals and comments to explain how the program works.
- Maintenance: Modifying and updating the program after implementation.

b) Explain functions of System Utilities.

Ans:

- Disk Management: Helps in formatting, partitioning, and organizing storage devices.
- File Management: Allows copying, moving, renaming, and deleting files.
- System Maintenance: Improves system performance by cleaning junk files and defragmenting disks.
- Security Utilities: Protects the system from viruses and unauthorized access.
- Backup and Recovery: Creates backups of data and restores data when needed.

c) List and explain the advantages of high level languages.

Ans:

- Easy to Learn and Use: Uses English-like statements that are simple to understand.
- Machine Independent: Programs can run on different computers with little or no modification.
- Reduced Development Time: Programs can be written faster with fewer lines of code.
- Easy Debugging and Maintenance: Errors are easier to find and programs are easy to modify.
- Rich Library Support: Provides built-in functions and libraries for faster development.

III PART - C

6.

a) State and prove De-Morgan's theorems for two variables.

Ans:

Proof of $(A + B)' = A'B'$

A	B	A + B	$(A + B)'$	A'	B'	$A'B'$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

Proof of $(A \cdot B)' = A' + B'$

A	B	A · B	$(A \cdot B)'$	A'	B'	$A' + B'$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

b) Write the procedure to perform r's complement subtraction with example.

Ans:

c) Using Venn diagram prove that $A \cdot (BC) = (AB) \cdot C$.

Ans: